

A SUPERSPACE APPROACH TO SUPERGRAVITY

SUPERSYMMETRY EXAM

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MOTIVATION

Local susy implies gravity.

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Global susy in terms of superfields.

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Global susy in terms of superfields.

Can we study local susy in superspace formalism?

JUST SAY 'SUPER'

CURVED SUPERSPACE

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- Supervielbein: $E_{\hat{M}}^N(z)$, $g_{\hat{M}\hat{N}} = (-)^{|\hat{M}|(|N|+|\hat{N}|)} E_{\hat{N}}^N E_{\hat{M}}^M \eta_{MN}$;

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- Torsion: $T_{\hat{M}\hat{N}}^N$. Curvature: $R_{\hat{M}\hat{N}M}^N$;
- $[\mathfrak{D}_M, \mathfrak{D}_N]_{|M||N|} = T_{MN}^P \mathfrak{D}_P - 1/2 R_{MN\underline{\alpha}\underline{\beta}} J^{\underline{\alpha}\underline{\beta}}$.

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- $\{\mathcal{D}_\alpha, \bar{\mathcal{D}}_{\dot{\alpha}}\} = -2i\sigma^\mu_{\alpha\dot{\alpha}} \partial_\mu.$

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Constraints on torsion tensor.

FOR YOUR SAFETY
LET'S SKIP THE (SUPER)COMPUTATION

Supergravity Algebra

$$\{\mathfrak{D}_\alpha, \mathfrak{D}_\beta\} = -1/2 R_{\alpha\beta\gamma\delta} J^{\delta\gamma};$$

$$\{\bar{\mathfrak{D}}_{\dot{\alpha}}, \bar{\mathfrak{D}}_{\dot{\beta}}\} = -1/2 R_{\dot{\alpha}\dot{\beta}\dot{\gamma}\dot{\delta}} J^{\dot{\delta}\dot{\gamma}};$$

$$\{\mathfrak{D}_\alpha, \bar{\mathfrak{D}}_{\dot{\alpha}}\} = T_{\alpha\dot{\alpha}}{}^\mu \mathfrak{D}_\mu - 1/2 R_{\alpha\dot{\alpha}\beta\gamma} J^{\gamma\beta} - 1/2 R_{\alpha\dot{\alpha}\dot{\beta}\dot{\gamma}} J^{\dot{\gamma}\dot{\beta}};$$

$$[\mathfrak{D}_\alpha, \mathfrak{D}_\mu] = T_{\alpha\mu}{}^\beta \mathfrak{D}_\beta - T_{\alpha\mu}{}^{\dot{\beta}} \bar{\mathfrak{D}}_{\dot{\beta}} - 1/2 R_{\alpha\mu\beta\gamma} J^{\gamma\beta} - 1/2 R_{\alpha\mu\dot{\beta}\dot{\gamma}} J^{\dot{\gamma}\dot{\beta}};$$

$$[\bar{\mathfrak{D}}_{\dot{\alpha}}, \mathfrak{D}_\mu] = T_{\dot{\alpha}\mu}{}^\beta \mathfrak{D}_\beta - T_{\dot{\alpha}\mu}{}^{\dot{\beta}} \bar{\mathfrak{D}}_{\dot{\beta}} - 1/2 R_{\dot{\alpha}\mu\beta\gamma} J^{\gamma\beta} - 1/2 R_{\dot{\alpha}\mu\dot{\beta}\dot{\gamma}} J^{\dot{\gamma}\dot{\beta}};$$

$$[\mathfrak{D}_\mu, \mathfrak{D}_\nu] = T_{\mu\nu}{}^\beta \mathfrak{D}_\beta - T_{\mu\nu}{}^{\dot{\beta}} \bar{\mathfrak{D}}_{\dot{\beta}} - 1/2 R_{\mu\nu\beta\gamma} J^{\gamma\beta} - 1/2 R_{\mu\nu\dot{\beta}\dot{\gamma}} J^{\dot{\gamma}\dot{\beta}}.$$

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Super-Weyl Transformation

$$\delta_\Sigma E_{\hat{M}}{}^\mu = (\Sigma + \Sigma^\dagger) E_{\hat{M}}{}^\mu;$$

$$\delta_\Sigma E_{\hat{M}}{}^\alpha = (2\Sigma^\dagger - \Sigma) E_{\hat{M}}{}^\alpha + i/2 E_{\hat{M}}{}^\mu (\bar{\mathfrak{D}}\Sigma^\dagger \bar{\sigma}_\mu)^\alpha;$$

$$\delta_\Sigma E_{\hat{M}\dot{\alpha}} = (2\Sigma - \Sigma^\dagger) E_{\hat{M}\dot{\alpha}} + i/2 E_{\hat{M}}{}^\mu (\mathfrak{D}\Sigma \sigma_\mu)_{\dot{\alpha}}.$$

Independent Superfields

- Chiral superfield $\mathcal{R}$;
- Real vector superfield $\mathcal{G}_\mu$;
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Supergravity Multiplet

- Massless graviton $e_{\hat{\mu}}^{\mu}$;
- Massless gravitino $(\psi_{\hat{\mu}}^{\alpha}, \bar{\psi}_{\hat{\mu}\dot{\alpha}})$;
- Auxiliary complex scalar M ;
- Auxiliary real vector b_{μ} .

CURVED SUPERSPACE: GAUGE FIXING

Using local Lorentz and superdiffeomorphisms, we can fix the gauge

$$\Omega_{\hat{\mu}M}^N| = \omega_{\hat{\mu}M}^N(x), \quad \Omega_{\hat{\alpha}M}^N| = 0, \quad \Omega_{\hat{\alpha}M}^{\hat{\alpha}N}| = 0;$$

$$E_{\hat{M}}^M| = \begin{pmatrix} e_{\hat{\mu}}^{\mu}(x) & 1/2 \psi_{\hat{\mu}}^{\alpha}(x) & 1/2 \bar{\psi}_{\hat{\mu}\dot{\alpha}}(x) \\ 0 & \delta_{\hat{\alpha}}^{\alpha} & 0 \\ 0 & 0 & \delta_{\hat{\alpha}}^{\dot{\alpha}} \end{pmatrix};$$

$$E_M^{\hat{M}}| = \begin{pmatrix} e_{\mu}^{\hat{\mu}}(x) & -1/2 \psi_{\mu}^{\hat{\alpha}}(x) & -1/2 \bar{\psi}_{\mu\hat{\alpha}}(x) \\ 0 & \delta_{\alpha}^{\hat{\alpha}} & 0 \\ 0 & 0 & \delta_{\dot{\alpha}}^{\hat{\dot{\alpha}}} \end{pmatrix};$$

$$\mathcal{R}(z)| = -1/6 M(x);$$

$$\mathcal{G}_{\mu}(z)| = -1/3 b_{\mu}(x).$$

CURVED SUPERSPACE: SUPERGRAVITY TRANSFORMATION

Defining $\tilde{\theta}_\alpha = \theta^{\hat{\alpha}} E_{\hat{\alpha}}{}^\beta \varepsilon_{\alpha\beta} \neq \theta_\alpha$, the supergravity transformation is

$$\begin{aligned}\xi^\mu(z) &= 2i(\tilde{\theta}\sigma^\mu\bar{\varepsilon} - \varepsilon\sigma^\mu\tilde{\bar{\theta}}), \quad \xi^\alpha(z) = \varepsilon^\alpha, \quad \bar{\xi}_{\dot{\alpha}}(z) = \bar{\varepsilon}_{\dot{\alpha}}, \\ L_{\alpha\beta}(z) &= 1/3 \left[\tilde{\theta}_\alpha(2\varepsilon_\beta M^* + b_{\beta\dot{\gamma}}\bar{\varepsilon}^{\dot{\gamma}}) + \tilde{\theta}_\beta(2\varepsilon_\alpha M^* + b_{\alpha\dot{\gamma}}\bar{\varepsilon}^{\dot{\gamma}}) \right], \\ L_{\dot{\alpha}\dot{\beta}}(z) &= 1/3 \left[\tilde{\bar{\theta}}_{\dot{\alpha}}(2\bar{\varepsilon}_{\dot{\beta}} M + \varepsilon^\gamma b_{\gamma\dot{\beta}}) + \tilde{\bar{\theta}}_{\dot{\beta}}(2\bar{\varepsilon}_{\dot{\alpha}} M + \varepsilon^\gamma b_{\gamma\dot{\alpha}}) \right], \\ L_{\mu\nu}(z) &= -1/2 \left[(\bar{\sigma}_\mu\sigma_\nu)^{\dot{\alpha}}{}_{\dot{\beta}} L_{\dot{\alpha}}{}^{\dot{\beta}} + (\sigma_\mu\bar{\sigma}_\nu)_\alpha{}^\beta L^\alpha{}_\beta \right],\end{aligned}$$

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The transformation of the supergravity multiplet is

$$\begin{aligned}\delta e_{\hat{\mu}}^\mu &= \delta E_{\hat{\mu}}^\mu \Big| = -i \left[\varepsilon(x) \sigma^\mu \bar{\psi}_{\hat{\mu}} - \psi_{\hat{\mu}} \sigma^\mu \bar{\varepsilon}(x) \right] \\ \delta \psi_{\hat{\mu}}^\alpha &= 2\delta E_{\hat{\mu}}^\alpha \Big| = -2\mathfrak{D}_{\hat{\mu}}\varepsilon^\alpha(x) + \frac{i}{3} e_{\hat{\mu}}^\nu (\bar{\varepsilon}(x) \bar{\sigma}_\nu)^\alpha M \\ &\quad + \frac{i}{3} e_{\hat{\mu}}^\nu [(\varepsilon(x) \sigma_\rho \bar{\sigma}_\nu)^\alpha b^\rho - 3\varepsilon^\alpha(x) b_\nu] \\ \delta M &= -6\delta\mathcal{R} \Big| = \dots \quad \delta b^\mu = -3\delta\mathcal{G}^\mu \Big| = \dots\end{aligned}$$

where $\mathfrak{D}_{\hat{\mu}}\psi_{\hat{\nu}}^\alpha = \partial_{\hat{\mu}}\psi_{\hat{\nu}}^\alpha + \psi_{\hat{\nu}}^\beta \omega_{\hat{\mu}\beta}^\alpha$.

SUPERFIELDS IN CURVED SPACETIME

Chiral Superfield: Definition

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- $\Phi = \Phi(z^{\hat{M}}) = \Phi(x^{\hat{\mu}}, \theta^{\hat{\alpha}}, \bar{\theta}_{\hat{\dot{\alpha}}});$
- Derivatives have Lorentz indices. Expansion has Einstein indices.

A NIGHTMARE

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- Components with Lorentz indices:

$$\begin{aligned}\phi &= \Phi|, & \xi_{\alpha} &= \frac{1}{\sqrt{2}} \mathfrak{D}_{\alpha} \Phi|, & F &= \frac{1}{4} \mathfrak{D} \cdot \mathfrak{D} \Phi|, \\ \phi^{\dagger} &= \Phi^{\dagger}|, & \bar{\xi}^{\dot{\alpha}} &= \frac{1}{\sqrt{2}} \bar{\mathfrak{D}}^{\dot{\alpha}} \Phi^{\dagger}|, & F^{\dagger} &= \frac{1}{4} \bar{\mathfrak{D}} \cdot \bar{\mathfrak{D}} \Phi^{\dagger}|;\end{aligned}$$

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- $\Xi^{\dagger} = [\mathcal{D} \cdot \mathcal{D} - 8\mathcal{R}^{\dagger}] \Phi, \quad \Xi = [\bar{\mathcal{D}} \cdot \bar{\mathcal{D}} - 8\mathcal{R}] \Phi^{\dagger}.$

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- Supergravity transformation:

$$\delta \Phi = -\xi^M \mathcal{D}_M \Phi = -\xi^{\alpha} \mathcal{D}_{\alpha} \Phi - \xi^{\mu} \mathcal{D}_{\mu} \Phi;$$

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- Field-strength superfield:

$$W_\alpha = -\frac{1}{4}(\bar{\mathfrak{D}} \cdot \bar{\mathfrak{D}} - 8\mathcal{R})e^{2gV}\mathfrak{D}_\alpha e^{-2gV}, \quad \bar{W}_{\dot{\alpha}} = \frac{1}{4}(\mathfrak{D} \cdot \mathfrak{D} - 8\mathcal{R}^\dagger)e^{-2gV}\bar{\mathfrak{D}}_{\dot{\alpha}} e^{2gV};$$

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$$W_\alpha = -\frac{1}{4}(\bar{\mathfrak{D}} \cdot \bar{\mathfrak{D}} - 8\mathcal{R})e^{2gV}\mathfrak{D}_\alpha e^{-2gV}, \quad \bar{W}_{\dot{\alpha}} = \frac{1}{4}(\mathfrak{D} \cdot \mathfrak{D} - 8\mathcal{R}^\dagger)e^{-2gV}\bar{\mathfrak{D}}_{\dot{\alpha}} e^{2gV};$$

- Under gauge: $W_\alpha \rightarrow e^{-2ig\Lambda} W_\alpha e^{2ig\Lambda}$;

Vector Superfield: Definition

- Definition: $V = V^\dagger$;
- Gauge transformation: $e^{2gV} \rightarrow e^{-2ig\Lambda} e^{2gV} e^{2ig\Lambda^\dagger}$;
- Wess-Zumino Gauge:

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- Under gauge: $W_\alpha \rightarrow e^{-2ig\Lambda} W_\alpha e^{2ig\Lambda}$;
- Physical components:

$$V_\mu = \frac{1}{4} \bar{\sigma}_\mu^{\dot{\alpha}\alpha} [\mathfrak{D}_\alpha, \bar{\mathfrak{D}}_{\dot{\alpha}}] V|, \quad \lambda_\alpha = \frac{i}{2g} W_\alpha|, \quad \bar{\lambda}_{\dot{\alpha}} = -\frac{i}{2g} \bar{W}_{\dot{\alpha}}|,$$

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- Supergravity transformation: $\delta V = -\xi^M \mathfrak{D}_M V$.

SUPERGRAVITY INVARIANT ACTION

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- Define $z^{\bar{M}} = (x^{\hat{\mu}}, \Theta^\alpha, \bar{\Theta}_{\dot{\alpha}})$.

WESS: 'A TOUR DE FORCE'

Defining Hybrid Variables

- Define $z^{\bar{M}} = (x^{\hat{\mu}}, \Theta^\alpha, \bar{\Theta}_{\dot{\alpha}})$ such that:

$$\begin{aligned}\Phi(x, \Theta) &= \Phi| + \sqrt{2}\Theta \cdot (\mathfrak{D}\Phi) \Big| - \frac{1}{4}\Theta \cdot \Theta (\mathfrak{D} \cdot \mathfrak{D}\Phi) \Big| \\ &= \phi + \sqrt{2}\Theta \cdot \chi - \Theta \cdot \Theta F;\end{aligned}$$

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- We can expand in Θ the superfields $\Xi, \Upsilon, W_\alpha, \mathcal{R}$.

Global Susy Action

$$\begin{aligned} S = & \int d^4x \, d^2\theta \, d^2\bar{\theta} \, \frac{1}{2} [K(\Phi, \Phi^\dagger e^{-2gV}) + K(e^{-2gV} \Phi, \Phi^\dagger)] \\ & + \int d^4x \, d^2\theta \, W(\Phi) + \int d^4x \, d^2\bar{\theta} \, W^*(\Phi^\dagger) \\ & + \frac{1}{16g^2} \int d^4x \, d^2\theta \, h_{ab}(\Phi) W^{a\alpha} W_\alpha^b + \frac{1}{16g^2} \int d^4x \, d^2\bar{\theta} \, h_{ab}^*(\Phi^\dagger) \bar{W}^{\dot{a}}_\alpha \bar{W}^{b\dot{\alpha}} \end{aligned}$$

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Principle of Covariance

- $\theta \rightarrow \Theta$;
- $d^2\theta \rightarrow d^2\Theta \mathcal{E}$;
- $\bar{\mathcal{D}} \cdot \bar{\mathcal{D}} \rightarrow (\bar{\mathfrak{D}} \cdot \bar{\mathfrak{D}} - 8\mathcal{R})$.

SUPERGRAVITY LAGRANGIAN: RESULT

Supergravity Lagrangian

$$\mathcal{L} = \int d^2\Theta \mathcal{E} \left\{ \frac{3}{8} (\bar{\mathfrak{D}} \cdot \bar{\mathfrak{D}} - 8\mathcal{R}) e^{-\frac{1}{3}K(\Phi, \Phi^\dagger e^{-2gV})} + W(\Phi) \right. \\ \left. + \frac{1}{16g^2} h_{ab}(\Phi) W^{a\alpha} W^b_{\alpha} \right\} + h.c.$$

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Units:

$$[\Theta] = [\theta] = -\frac{1}{2}, [\mathcal{E}] = [e] = 0, [\mathfrak{D}] = [\mathcal{D}] = \frac{1}{2}, [K] = 2.$$

Expanding:

$$\mathcal{L} \supset \frac{3}{8} M_P^2 \int d^2\Theta \mathcal{E} (\bar{\mathfrak{D}} \cdot \bar{\mathfrak{D}} - 8\mathcal{R}) e^{-\frac{1}{3M_P^2} K(\Phi, \Phi^\dagger e^{-2gV})} + h.c.$$

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Super-Weyl Transformation

$$\begin{aligned}\mathcal{L} \rightarrow \mathcal{L}' = & \frac{3}{8} \int d^2\Theta \mathcal{E}(\bar{\mathfrak{D}} \cdot \bar{\mathfrak{D}} - 8\mathcal{R}) e^{2(\Sigma + \Sigma^\dagger)} \exp \left\{ -\frac{1}{3} K(\Phi, \Phi^\dagger e^{-2gV}) \right\} \\ & + \int d^2\Theta \mathcal{E} e^{6\Sigma} W(\Phi) + \frac{1}{16g^2} \int d^2\Theta \mathcal{E} h_{ab}(\Phi) W^{a\alpha} W_\alpha^b + h.c.\end{aligned}$$

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Kahler Transformation

To have an invariant scalar potential:

$$K(\Phi, \Phi^\dagger e^{-2gV}) \rightarrow K(\Phi, \Phi^\dagger e^{-2gV}) + F(\Phi) + F^*(\Phi^\dagger), \\ W(\Phi) \rightarrow e^{-F(\Phi)} W(\Phi).$$

SUPERGRAVITY LAGRANGIAN: SYMMETRIES

Super-Weyl Transformation

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Then:

$$\begin{aligned}\mathcal{L}' \rightarrow \mathcal{L}'' = & \frac{3}{8} \int d^2\Theta \mathcal{E}(\bar{\mathfrak{D}} \cdot \bar{\mathfrak{D}} - 8\mathcal{R}) e^{2(\Sigma + \Sigma^\dagger)} e^{-\frac{1}{3}K - \frac{1}{3}(F + F^*)} \\ & + \int d^2\Theta \mathcal{E} e^{-F + 6\Sigma} W + \frac{1}{16g^2} \int d^2\Theta \mathcal{E} h_{ab}(\Phi) W^{a\alpha} W_\alpha^b + h.c.\end{aligned}$$

Generalised Kahler Potential

- For $F = \ln(W)$:

$$\begin{aligned}\mathcal{L} = & \frac{3}{8} \int d^2\Theta \mathcal{E} (\bar{\mathfrak{D}} \cdot \bar{\mathfrak{D}} - 8\mathcal{R}) e^{-\frac{1}{3}\mathcal{G}(\Phi, \Phi^\dagger e^{-2gV})} + \int d^2\Theta \mathcal{E} \\ & + \frac{1}{16g^2} \int d^2\Theta \mathcal{E} h_{ab}(\Phi) W^{a\alpha} W_\alpha^b + h.c.\end{aligned}$$

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- Generalised Kahler potential: $\mathcal{G} = K + \ln |W|^2$

Final Result

$$\begin{aligned}\mathcal{L} = & \frac{3}{8} \int d^2\Theta \mathcal{E}(\bar{\mathfrak{D}} \cdot \bar{\mathfrak{D}} - 8\mathcal{R}) e^{-\frac{1}{3}\mathcal{G}(\Phi, \Phi^\dagger e^{-2gV})} + \int d^2\Theta \mathcal{E} \\ & + \frac{1}{16g^2} \int d^2\Theta \mathcal{E} h_{ab}(\Phi) W^{a\alpha} W_\alpha^b + h.c.\end{aligned}$$

EXPLICIT COMPUTATION

HAVE FUN
COMPUTING THE COMPONENTS